

PET final-design selection study

Summary (generated from committed results)

PET final-design study

PET is diagnostic method development. Nothing here is a publication adoption; no historical threshold, verdict or the predecessor campaign's disposition changes; simulation only.

The question, and how this talk is organized

Question. Can a stronger PET — borrowing parts of Gregor's PET2 work — give an accurate, robust and calibrated unfolding of MINERvA E_{avail} , and how does it compare with scalar GBDT OmniFold?

1. **What we tried**, and how it relates to Gregor's work.
2. **How we set it up**: what we measure, and the decision rules, fixed before the results.
3. **Results**: development, the two finalists, uncertainty calibration, PET vs GBDT.
4. **What the results mean** for the decisions.

Signal-only simulation throughout. PET is diagnostic method development, not a publication input.

1. Where we started: our PET and Gregor's PET2 on the same test

- ▶ **Earlier comparison (September 2026).** Our production PET recipe and Gregor's PET2-small arm (his backbone, his 33-token typed inputs, his pretrained checkpoint, adapted to MINERvA) unfolded the same injected E_{avail} tilt, 8 seeds each, 3 iterations.
- ▶ **Result.** Recovery: Gregor's arm 0.417, ours 0.304 (his better by 0.113). Both fell below the absolute floor 0.556 (80 % of a pinned achievable-recovery ceiling): NEITHER ELIGIBLE, no selection.
- ▶ **What that comparison said to borrow** from his setup: event-level energy sums with an overflow token, a particle-type embedding, one capacity step — and, as whole designs, his backbone and checkpoint.
- ▶ **This study:** build complete designs from those parts plus a stronger truth step of our own, and test them against rules fixed in advance.

1. Design names

name part	meaning
CTL	historical production PET recipe: misses carried, 3 iterations, 8 epochs, forced low rate after iteration 0
C	CTL with the predecessor's efficiency-corrected truth step
B	predecessor's feature arm: reco energy summaries at step 1, one-hot particle type at step 2
H1	C + categorical truth particle type (13-class one-hot) in the truth step
H2	H1 + reco summaries at the detector step (reco E_{avail} , reco q_3 , cluster ΣE and count)
S1	constant learning rate at every iteration
E16 / T24	16 epochs per fit at both steps / 24 epochs per truth-step fit
L64 / L128	wider detector-step PET (projection 64 / 128, 4 layers)
P2pre / P2scr	Gregor's PET2-small backbone at the detector step on its 33-token typed inputs, pretrained / from scratch
A1	annealed detector-step learning rate after iteration 1
X4	4 independently seeded truth-step fits averaged per iteration
$K=5$, $K=4$	the frozen number of OmniFold iterations

Names compose: $H2S1T24 = H2 + S1 + T24$. Step 1 reweights simulation to match (pseudo)data at detector level; step 2 turns those weights into truth-level weights; one pass is one iteration.

1. Designs tried, and what came from Gregor's work

design	what it changes	from Gregor's work
H1	C + categorical truth PDG (13-way one-hot)	type encoding (cmp 2), padding kept distinct; truth step is ours
H2	H1 + reco summaries at step 1	whole-event energy sums + overflow (cmp 4, 6; not per-family)
CS1	C + constant rate	—
H2S1	H2 + constant rate	as H2
H2S1E16	H2S1, 16 epochs	as H2
L64H2	H2, step-1 PET width 64	as H2 + one capacity step (cmp 7)
L128H2	H2, step-1 PET width 128	as H2 + one capacity step (cmp 7)
L64S1	H2S1, width 64	as L64H2
L128S1	H2S1, width 128	as L128H2
L128S1E16	L128S1, 16 epochs	as L128H2
P2preS1	PET2-small at step 1, pretrained	his backbone, checkpoint, 33-token typed inputs, recipe
P2scrS1	PET2-small, random init	as P2preS1 without the checkpoint
P2preA1	P2preS1, annealed step-1 rate	as P2preS1
P2scrA1	P2scrS1, annealed step-1 rate	as P2scrS1
H2S1T24	H2S1, 24-epoch truth step	as H2
L128S1T24	L128S1, 24-epoch truth step	as L128H2

Anchors, not selectable: CTL (historical production recipe), C (CTL with an efficiency-corrected truth step), B (predecessor's feature arm). Gregor-derived parts follow the pinned configuration comparison (cmp sections); the truth-step design is the study's own.

2. How the numbers are measured

- ▶ **Simulation only.** Pseudodata are simulated events with a known, injected change to the truth (e.g. an E_{avail} tilt, more protons); the unfolding starts from the unchanged simulation (the prior).
- ▶ **Recovery** $R = 1 - L_1(\text{unfolded} - \text{target}) / L_1(\text{prior} - \text{target})$ on normalized histograms: 1 = the injected change fully recovered, 0 = no better than the prior, < 0 = moved away.
- ▶ **Tests:** energy tilt and its opposite (E0, E3), proton topology (E4, $E_{\text{avail}} \times$ proton class), energy $\times$ momentum transfer (E5), a 21-case robustness library, weight tails and seed spread.
- ▶ **Fresh events:** designs were developed on a DEV bank; the final test used a bank no earlier work had scored, blinded until every final run was complete. A third bank stays sealed.
- ▶ **Rules frozen first:** every threshold, finalist and sample size was committed before final results were seen; the outcome is whatever those rules give.
- ▶ **Coverage:** does the six-member bootstrap interval contain the true value at its nominal rate across 120 independent replicates?

2. Decision rules (frozen in advance)

1. **Eligibility first.** A design must pass every rule at its own frozen settings:
 - *accuracy*: recovery floors on the energy tilt, its opposite, the regions, proton topology and energy $\times q_3$, at a simultaneous 95 % lower bound;
 - *robustness*: across a 21-case library it rarely moves away from the truth;
 - *stability*: no extreme weights, and a repeat run with new seeds agrees to within 0.05;
 - *calibration*: its uncertainty intervals cover the truth at their nominal rate.
2. **Then choose.** The smaller network wins only if it is as accurate (within 0.02) and at least twice as cheap; otherwise the stronger, larger design.
3. **No eligible design means no winner** — and no default.

Every threshold, finalist and sample size was frozen before any final-bank result was seen.

3. Development: which designs survived

design	development result (DEV bank)
H1	S-U4 fails at every k (best 0.023; need ≥ 0.25)
H2	no k passes all four screens
CS1	S-N2 seed sd 0.054 > 0.05
H2S1	S-N2 seed sd 0.095 > 0.05
H2S1E16	S-N2 seed sd 0.074 > 0.05
L64H2	no k passes all four screens
L128H2	no k passes all four screens
L64S1	S-N2 seed sd 0.057 > 0.05
L128S1	S-N2 seed sd 0.091 > 0.05
L128S1E16	S-N2 seed sd 0.071 > 0.05
P2preS1	S-N1 step-1 weight tail at $K^* = 4$
P2scrS1	S-U4 fails at every k (best 0.110; need ≥ 0.25)
P2preA1	S-U4 fails at every k (best 0.225; need ≥ 0.25)
P2scrA1	S-U4 fails at every k (best 0.060; need ≥ 0.25)
H2S1T24	passed every development screen: finalist
L128S1T24	passed every development screen: finalist

Screens: S-U1 tilt, S-U3 opposite tilt, S-U4 proton topology, S-B2 D4d residual; then S-N1 weight tail and S-N2 seed spread. The step-2 ensemble fallback (H2S1X4, L128S1X4) never met its entry condition. The finalists then went to the fresh final bank (next slides).

3. The two finalists side by side

quantity (final bank, look 1)	H2S1T24 $K=5$	L128S1T24 $K=4$
Energy-tilt recovery (E0)	0.895	0.863
Proton-topology recovery (E4)	0.391	0.348
Joint energy-momentum-transfer recovery (E5)	0.446	0.450
Estimator-seed spread of energy recovery (N2)	0.037	0.043
Largest observed truth weight (N1)	90.2	33.4
Measured A100-hours per unfolding	1.77	1.52
Frozen point-estimator rules	all pass	B2 fails
Interval calibration (coverage)	fails (C1, C4)	not tested (ineligible on B2)

Recoveries: 1 = exact, 0 = no better than the prior. B2's failure is a point decision whose interval crosses the limit; the two designs are not shown to differ in robustness.

3. Uncertainty calibration: coverage of H2S1T24

H2S1T24 $K=5$, $B = 6$, 120 replicates	measured	verdict
C1 pooled 95 % (LB ≥ 0.90 , point ≤ 0.99)	0.990 (LB 0.980)	unresolved
C1 pooled 68 % (LB ≥ 0.60 , point ≤ 0.80)	0.896 (LB 0.858)	decisive
C1 (both levels)		FAIL
C2 every bin 95 % ≥ 0.85	all bins pass	PASS
C3 moderate / good 95 % LB ≥ 0.85	0.949 / 0.993	PASS
C4 half-width / limit, bins 1–6 (≤ 1)	1.32–1.72	FAIL
C5 D4c up	skipped after decisive C1–C4 FAIL	—

In aggregate the six-member interval is **too wide**; the same against the FB population target. Not uniformly conservative: the low-acceptance region **under-covers** (68 %: 0.26, 95 %: 0.79), a region no rule gates. Mechanism not established.

3. PET vs GBDT OmniFold on the same events

paired difference in R , PET – GBDT	H2S1T24 $K=5$	L128S1T24 $K=4$
energy tilt E0 (60 draws)	+0.102 [0.084, 0.119]	+0.070 [0.051, 0.089]
opposite tilt E3 (60)	+0.076 [0.059, 0.093]	+0.070 [0.053, 0.088]
proton topology E4 (40)	+0.055 [0.035, 0.075]	+0.012 [−0.007, 0.031]
E4 against GBDT run to $k=10$	+0.009 [−0.012, 0.030]	−0.034 [−0.054, −0.015]
energy $\times q_3$ E5 (40)	+0.009 [−0.009, 0.026]	+0.013 [−0.005, 0.030]
generator reweighting: NuWro (8)	−0.247 [−0.345, −0.148]	−0.208 [−0.324, −0.092]
NuWro' (8)	−0.216 [−0.303, −0.130]	−0.180 [−0.241, −0.118]
GiBUU (8)	−0.206 [−0.305, −0.107]	−0.121 [−0.177, −0.065]

same draws	H2S1T24	L128S1T24	GBDT
B2 residual, D4d n down (PET rule ≤ 0.010)	0.0097	0.0122	0.0033
largest truth weight	90.2	33.4	5.6
GBDT's E0 summed per-bin MSE relative to PET	$2.9\times$	$1.9\times$	1

GBDT = the matched study's tuned scalar OmniFold (HistGradientBoosting, efficiency-corrected, reco summaries + muon kinematics, truth with species counts, $k=7$ chosen on DEV), run on the same look-1 final-bank units with the same events, targets and scorer. Unadjusted 95% intervals; exploratory. GBDT error is mostly systematic under-correction. No uncertainty comparison: this GBDT has no interval procedure here. On DEV, binned IBU and AUSSIE match PET on the 1D tilt and fail hadron or generator cases.

4. What the evidence supports

question	what the evidence supports
Can improved PET recover meaningful physics changes? Is it more accurate than GBDT OmniFold?	Yes, in this simulation test. H2S1T24 $K=5$ passed every frozen point-estimator rule: energy tilt 0.895, proton topology 0.391, energy $\times q_3$ 0.446. On energy shape, yes; on generator changes, no. On the same final-bank events PET beats a tuned scalar GBDT OmniFold on the energy tilt (H2S1T24 +0.102, L128S1T24 +0.070 in R) but is worse on every generator-model reweighting (NuWro -0.247, GiBUU -0.206 for H2S1T24). Binned IBU (0.923 on the development tilt) matches PET on the 1D tilt but fails the proton case.
Can it give tighter, reliable confidence intervals?	Unestablished. Its six-member bootstrap interval failed calibration: too wide overall (68 % intervals cover 0.896), too narrow at low acceptance (0.257).
Will it distinguish generators more strongly?	Unestablished. That needs calibrated inference including detector and model uncertainties; no tested procedure provides it, and PET recovers generator-model changes less well than GBDT.

Terminal outcome by the frozen rules: **NO_ELIGIBLE_DESIGN**, no selected default. Signal-only simulation, conditional on the frozen event banks; PET is diagnostic.

4. What this means for the decisions

- ▶ **Outcome by the frozen rules: NO_ELIGIBLE_DESIGN.** No PET design is selected and no default is named. PET stays diagnostic; the GBDT-based production analysis is unchanged.
- ▶ **What PET would buy:** better energy-shape and hadron-species recovery than this GBDT on the same events. **Its main risk:** generator-model dependence, exactly what an uncertainty has to cover; and no calibrated interval exists yet.
- ▶ **More study-scale comparison would not change the next decision:** ranking the two finalists picks between near-equal designs, and calibrating at study scale does not transfer to the data scale.
- ▶ **If PET is pursued (owner decisions, none authorized):**
 - a data-scale readiness step (P0, capped at 165 A100-h): closures on the development tilt, NuWro and null with matched GBDT and IBU baselines, plus two diagnostic real-data runs; it stops with a no-go if PET does not beat GBDT there. Estimated cost of one data-scale unfolding: 7.3 (3.7–14.6) A100-h at a 2 M-event prior, scaled from the study;
 - or the study's own repair: an interval calibrated on DEV and re-validated on the sealed bank.

Appendix

Backup

Reference methods and alternatives

method	what it is	from Gregor's work	result
CTL $K=3$	historical production PET recipe (anchor)	—	final bank energy tilt 0.306 (floor 0.556)
C $K=3$	efficiency-corrected truth step (anchor)	—	final bank energy tilt 0.560
B	predecessor's feature arm (control)	energy sums (cmp 4)	development tilt 0.474 at $k=3$
Gregor's arm	his PET2-small, his tokens, pretrained; historical comparison, $k=3$	his configuration (adapted)	0.417 vs ours 0.304; floor 0.556: NEITHER ELIGIBLE
GBDT OmniFold	scalar HistGradientBoosting on reco p_T , $p_{ }$, E_{avail} , q_3 , cluster sums	—	development tilt 0.783 (efficiency-corrected) / 0.418 (misses carried), $k=3$; proton topology 0.389 only with species-count truth inputs
binned IBU	response-matrix unfolding, efficiency-corrected	—	development tilt 0.923; moves away in the proton case (-0.381)
AUSSIE	scalar method of arXiv:2602.24282, matched to GBDT OmniFold	—	lost all 12 matched comparisons on robustness and stability

Anchors are not selectable; scalar references use the predecessor's DEV-bank selections, so they are context, not matched comparisons with the PET finalists.

Algorithmic alternative: AUSSIE (scalar, matched)

AUSSIE setting vs OmniFold $k=3$	moves-away: AUSSIE	OmniFold	worst R : AUSSIE	OmniFold
$\lambda=1000$, misses carried, kinematics	11	6	-1.077	-0.217
$\lambda=1000$, misses carried, kinematics + species	8	6	-0.985	-0.230
$\lambda=0$, efficiency-corrected, kinematics	14	6	-3.876	-0.392
$\lambda=0$, efficiency-corrected, kinematics + species	7	6	-0.735	-0.352

Bounded matched stress test (same scalar inputs, miss handling, data and tuning opportunity); truth kinematics = $(E_{\text{avail}}, p_T, p_{\parallel}, q_3)$. 12 comparisons: 2 miss rules $\times$ 2 truth-input sets $\times$ 3 OmniFold operating points (shown: $k=3$). AUSSIE wins none of the 12 on robustness or stability, so it does not advance to a PET backbone.